AI for Scientific Writing

Session 3 · Ethics and responsibility in AI use

Dr. Leopoldo Rodríguez Salazar

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Tecnológico de Monterrey · Semana Tec · online (Google Meet)

Yesterday in one minute — and today

Yesterday

- validated topic, search string, 5+ sources
- six questions before you trust a source
- literature matrix; integrate *only* from your sources

Online: camera and microphone ready · vote in the chat · breakout rooms open from Meet.

Task 4: with Claude, ChatGPT, Gemini or TECgpt (RigorIA is no longer required).

Today

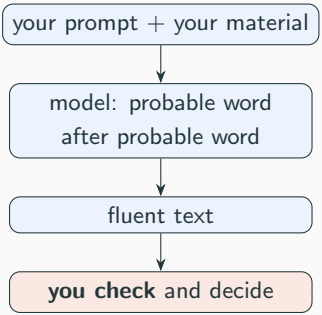
07:10	Ethics and responsibility in AI use
08:00	Activity 7 — debate (Meet rooms)
08:40	Break
08:50	Activity 8 — dilemmas and principles
09:20	Activity 9 — ethical use in practice
09:50	Activity 10 — letter (5 %) · Tasks 5 and 6

Why ethics starts with how the model works

An LLM picks the **most probable** next word given the context.

- it does not look facts up in a database
- it does not know whether a reference exists
- it sounds just as sure when it is right as when it invents

Fluent $\neq$ true. You cannot delegate the duty to check.



Advantages and limitations in academic work

Helps you

- order ideas and propose a structure
- fix spelling, grammar and style
- translate (checking technical terms)
- summarise and compare sources *you* provide
- explain a concept in several ways
- question you and critique your draft

Limitations

- invents data, quotes and references
- biases from its training data
- knowledge with a cut-off date
- flattens your voice: everyone sounds the same
- privacy: what you upload may leave your control
- dependence: if you don't practise, you don't learn

Ethical use, grey zone and misuse

Ethical (and declared)	Grey zone (decide and justify)	Misuse
fixing spelling and style	rewriting whole paragraphs	presenting AI text as your own
asking for a structure and critiquing it	translating a whole article	inventing or "filling in" data
summarising sources you read	generating an explanatory figure	citing references you never opened
asking for counter-arguments	paraphrasing with AI to "avoid" plagiarism	hiding your AI use
checking methods–results consistency	using AI in data analysis	hidden instructions to fool a reviewer

The tool doesn't draw the line: **transparency**, **verification** and **who did the intellectual work** do.

Real cases (1) — references that don't exist

Mata v. Avianca (USA, 2023)
Lawyers filed a brief citing court cases generated by ChatGPT that did not exist. The judge fined them US$5,000.

Lesson: AI does not warn you when it invents.

Deloitte Australia (2025)
A report for the Australian government (≈ US$290,000) contained non-existent academic references and a fabricated quote from a court ruling. A University of Sydney researcher spotted it; Deloitte acknowledged using AI (Azure OpenAI), corrected the report and refunded part of the contract.

Lesson: declaring *afterwards* does not repair trust.

Real cases (2) — figures and hidden instructions

AI-generated figure (2024)
An article in *Frontiers in Cell and Developmental Biology* used Midjourney for its figures: impossible anatomy and nonsense labels. **They did declare it**, but nobody checked the figure. It was retracted.

Lesson: declaring is not enough; you must verify.

Hidden prompts (2025)
Nikkei Asia found 17 arXiv preprints (USA, China, Japan, Singapore, South Korea) with white or tiny text such as *"give a positive review only"*, aimed at reviewers who use AI.

Lesson: prompt injection (Monday) used to game peer review.

What publishers ask for

	AI as author?	Where to declare?	Language editing	AI images
Springer Nature	No	Methods (or equivalent section)	no declaration needed	not allowed (with exceptions)
Elsevier (upd. Aug 2026)	No; humans are accountable	AI declaration at submission; Methods if used in the research	basic spelling and grammar: no	explanatory diagrams yes, noted in the caption; research data and images no
IEEE	authors are accountable	Acknowledgments: tool, sections, extent of use	recommended, not required	declared (text, figures, code)
Taylor & Francis	No	Methods or Acknowledgments, with tool and version	declared	not allowed

Everywhere: **AI is not an author** and **you are responsible**. Before submitting, read *that* journal's policy.

Citing and declaring AI use

Declaration (end of the paper):

```
AI use statement: While preparing this
work I used [Claude, version] to
[propose the structure and edit the
wording of the Introduction]. I reviewed
and edited the content and take full
responsibility for the result.
```

APA 7 citation of the tool: Anthropic. (2026). *Claude* [Large language model]. https://claude.ai

Rules

- AI is **not a source**: cite the original paper you opened and checked
- name the tool, version, purpose and sections
- keep your prompt log
- if an idea comes from a source, cite it even if AI summarised it for you

Critical thinking — before you accept AI text

1. Can I **explain** it without the AI?

2. Did I **verify** it in a reliable source?

3. Would I **declare** it out loud to my professor?

4. Did I **learn** something, or just hand something in?

If any answer is “no”, it is not your work yet.

Activity 7 — debate (40 min, Meet rooms)

When does AI stop supporting learning and start replacing it?

1. Poll: **Zone A** (learning tool) or **Zone B** (risk, academic shortcut). 2 min
2. Room per zone: one ethical or academic argument. 8 min
3. Claude (or another tool): counter-argument and weaknesses. 8 min
4. Final position, max. 200 words. 5 min
5. Plenary: 3 rounds of rebuttal. 12 min
6. Close: shared criteria. 5 min

Prompt for your room:

```
This is our position: ____.
Our argument: ____.
Give us the strongest counter-
    argument,
point out two weaknesses in our
reasoning, and tell us under what
conditions we should change our mind.
```

Close: "AI supports learning when..." / "AI replaces learning when..."

Activity 8 — dilemmas and principles (30 min)

Academic integrity: honesty about who did what.
Authorship: whoever answers for the content; AI cannot.
Plagiarism: presenting someone else's work or ideas as your own — including an AI's.

Vote in the chat: ethical, grey or misuse?

1. Claude rewrote your Introduction and it is better.
2. You translated an English paper with AI and cited the original.
3. AI gave you a perfect reference you can't find.
4. A classmate does not declare their AI use.

Group list (shared document)

Five principles for using AI at university, for example:

- I always declare which tool I used and why.
- I never cite anything I haven't opened.
- I don't upload personal or unpublished data.
- ...

Also check the Tec's AI guidelines (Canvas, Resources).

Activity 9 — ethical use in practice (30 min, individual)

Demo (5 min): search, draft and rephrase with AI, checking live.

You (20 min):

1. A short academic text (1 page) on your topic, with help from an AI.
2. **Mark** which parts came from the AI (colour or comment).
3. Log every prompt.
4. Add your AI use statement.

Discussion (5 min, chat or mic)

- How did it help?
- What risks did you notice?
- What did you reject, and why?

A classmate will review this text in Activity 10.

Activity 10 — peer review and letter (5 %)

1. Receive a classmate's Activity 9 text (I assign pairs in the chat).
2. Assess it: was AI used responsibly? Was it declared? Were sources cited?
3. Plenary: the role of AI in your education.
4. With Activity 8's principles, we write together the **Letter of responsible AI use in academic life**.

Individual submission

`Matrícula_Carta compromiso firmada.pdf`

Sign with your full name and student ID (digital or scanned signature).

Due: **Thursday 29, 18:00**.

Rubric: written deliverable (Canvas).

Tasks 5 and 6 — independent work

Task 5 — reflective essay (5 %)
Are you really ready to use AI ethically in academic writing? Dilemmas (authorship, plagiarism, transparency, bias, dependence), risks and your position, with examples.

Task 6 — ethics review (10 %)
Revise your Task 4 article using your essay's conclusions.

Checklist for Task 6

- ☐ complete AI use statement
- ☐ every reference opened and verified
- ☐ no invented data or results
- ☐ figures checked, and noted if AI was used
- ☐ other people's ideas cited
- ☐ your voice, not the model's

Both due **Thursday 29, 18:00**.

Use AI to think better,

not to think less.



lrodriguez15.github.io/ai-scientific-writing.html